

Draft Lesson Plan – Lending Kit Option #1

Middle School Bike, E-Bike & Pedestrian Safety Education

Grade Level: 6–8

Time: 60–75 minutes

Learning Objectives

Students will be able to:

- Explain basic bicycle, e-bike, and pedestrian safety rules.
- Properly fit and wear a bicycle helmet.
- Perform an ABC Quick Check before riding.
- Demonstrate correct bicycle hand signals.
- Identify common traffic signs and understand right-of-way rules.
- Practice safe behaviors when sharing roads, sidewalks, crosswalks, bike lanes, and multi-use paths.
- Explain the differences between traditional bicycles and e-bikes.
- Describe how speed affects stopping distance and reaction time while riding an e-bike.
- Demonstrate responsible decision-making while operating or riding around e-bikes.
- Recognize basic e-bike battery safety and proper charging practices.

Materials

- Bicycle helmets
- Bicycle(s)
- E-bike (or photos/video if one is unavailable)
- **ABC Quick Check inspection cards**
- Traffic sign and signal cards
- Safety cones
- Hand signal reference cards
- Sidewalk chalk or roadway tape
- Reflective vests or stickers
- E-bike classification poster
- **Speed comparison chart (Bike vs. E-bike)**
- **"Safe or Unsafe?" scenario cards**
- **Student worksheets**
- **Exit tickets**

Lesson Procedures

Introduction (10 minutes)

- Begin by asking students how they usually travel to school.
- Discuss the importance of bicycle, e-bike, and pedestrian safety.
- Review the lesson objectives and explain why sharing the road safely benefits everyone.

Activity 1: Helmet Fit & ABC Quick Check (15 minutes)

- Demonstrate how to properly fit a bicycle helmet.
- Introduce the ABC Quick Check (Air, Brakes, Chain, Quick Releases).
- Have students practice inspecting a bicycle using the checklist.

Activity 2: Rules of the Road & Shared Spaces (15 minutes)

- Review traffic signs
- Right-of-way
- Hand signals
- Discuss e-bike classes
- Why speed changes stopping distance
- Why sidewalks aren't always appropriate
- Battery charging safety
- Wearing helmets on e-bikes

Activity 3: Bike Skills Course (20 minutes)

Students rotate through stations that include:

- Helmet fit check
- Hand signal practice
- Obstacle course using cones
- Traffic sign recognition
- Safe pedestrian crossing demonstration
- E-Bike Decision Station

Conclusion (10 minutes)

- Review the key safety concepts learned during the lesson.
- Ask students to identify one new safety habit they will practice.
- Complete a short exit ticket or worksheet to assess understanding.

Assessment

- Participation in class discussions and activities.
- Successful completion of the ABC Quick Check.
- Demonstration of correct hand signals.
- Completion of the exit ticket or reflection worksheet.

Condensed Sample Activities

- Helmet Fit & ABC Quick Check
- Traffic Signs & Hand Signals
- Bike Skills Course (Bike Rodeo)
- E-Bike & Shared Path Safety
- Safe Route Planning Activity
- Battery Safety Discussion

Discussion questions

- "What makes an e-bike different from a regular bike?"
- "Why might an e-bike take longer to stop?"
- "Would you ride an e-bike on a crowded sidewalk? Why?"
- "What could happen if you're looking at your phone while riding?"

Exit Tickets?

- What's one thing you'll always check before riding?
- Name one difference between riding a bike and riding an e-bike.

Introduction

This lesson was developed as part of the Bike and Pedestrian Safety Kit Loan Program to provide schools, parks, and community organizations with an easy-to-use lesson that can be taught by a variety of instructors. The lesson is designed to be interactive, engaging, and practical while helping middle school students develop the knowledge and skills needed to safely walk, ride bicycles, and ride e-bikes.

This lesson is intended for students in grades 6–8 because students at this age are becoming more independent in the ways they travel. Many students walk, ride bicycles, ride e-bikes, or use other forms of active transportation to get to school or around their neighborhoods. As their independence increases, it is important that they understand how to safely share roads, sidewalks, bike lanes, crosswalks, and other public spaces with pedestrians, cyclists, and motorists.

Rather than relying on lecture alone, this lesson emphasizes hands-on learning through demonstrations, discussions, and skill-building activities. Students will learn how to properly fit and wear a bicycle helmet, perform an ABC Quick Check before riding, recognize common traffic signs and signals, demonstrate correct bicycle hand signals, and identify safe behaviors when riding bicycles or e-bikes. Students will also learn about the differences between traditional bicycles and e-bikes, including how increased speed affects stopping distance, why riders should slow down around pedestrians, and the importance of responsible riding and battery safety.

To allow students adequate time to both learn and practice these skills, the lesson has been divided into two separate lessons. Lesson 1 takes place in the classroom and focuses on bicycle, pedestrian, and e-bike safety concepts. Lesson 2 takes place outdoors and provides students with the opportunity to apply what they learned by participating in a Bike Skills Course (Bike Rodeo).

Lesson 1: Classroom Instruction

Time: 60–75 Minutes

Introduction (10 Minutes)

Teacher Script

"Good morning, everyone! Before we begin today's lesson, I'd like to ask you a question. How do you usually travel to school? Raise your hand if you walk, ride a bicycle, ride an e-bike, ride a scooter, or get dropped off by someone."

(Allow students time to respond.)

"Even if you don't ride a bicycle every day, everyone is a pedestrian at some point. Whether you're walking to school, crossing the street, or riding through your neighborhood, knowing how to stay safe is important."

"Today we're going to learn about bicycle safety, pedestrian safety, and e-bike safety. By the end of today's lesson, you'll know how to properly wear a helmet, inspect a bicycle before riding, recognize important traffic signs, communicate with drivers using hand signals, and understand why e-bikes require extra responsibility."

Activity 1: Helmet Fit & ABC Quick Check (20 Minutes)

Teacher Script

"Before anyone rides a bicycle, the first thing they should think about isn't speed—it's safety."

(Hold up a bicycle helmet.)

"Can anyone tell me why wearing a helmet is important?"

(Allow students to answer.)

"A properly fitted helmet can greatly reduce the risk of a serious head injury, but it only works if it's worn correctly. An easy way to remember how to fit a helmet is the **2-2-2 Rule**."

"The first **2** means the helmet should sit about **two fingers above your eyebrows** so it protects your forehead."

"The second **2** means you should be able to fit **two fingers between your chin and the chin strap**. It should be snug, but still comfortable."

"The last **2** reminds us that the straps should form a **V shape under each ear**, with the buckle resting just below the earlobe."

"Your helmet should also sit level on your head and shouldn't rock forward or backward when you move."

"Now it's your turn. Pair up with a partner and use the 2-2-2 Rule to check each other's helmet fit before we move on."

After students have practiced helmet fitting, transition into the bicycle inspection.

"Before every ride, it's important to make sure your bicycle is safe to ride. One easy way to remember this is by using the ABC Quick Check."

Explain each part of the ABC Quick Check:

- **A – Air:** Check that the tires are properly inflated.
- **B – Brakes:** Make sure both brakes work correctly.
- **C – Chain:** Inspect the chain to make sure it is clean and working properly.

Demonstrate each step on a bicycle before allowing students to complete the ABC Quick Check in pairs using the provided inspection cards.

Activity 2: Rules of the Road, E-Bike Safety & Shared Spaces (25 Minutes)

Teacher Script

"Knowing how to ride a bicycle is only part of being a safe rider. It's also important to understand the rules of the road."

Display several traffic signs one at a time and ask students to identify what each sign means. Discuss stop signs, yield signs, crosswalks, bike lanes, and traffic signals.

Next, introduce bicycle hand signals.

"When you're riding on the road, drivers can't read your mind. Hand signals help communicate where you're going."

Demonstrate the signals for left turn, right turn, and stopping before allowing students to practice each signal together.

After reviewing hand signals, transition into e-bike safety.

"Now let's talk about e-bikes. Even though they look very similar to regular bicycles, there are some important differences."

"Did you know there are three different classes of e-bikes? Each class works a little differently."

Class 1

"A Class 1 e-bike only provides assistance while you're pedaling, and the motor stops helping once you reach 20 miles per hour."

Class 2

"A Class 2 e-bike also has a throttle, which means the bike can move without pedaling. Like Class 1, the motor stops assisting at 20 miles per hour."

Class 3

"A Class 3 e-bike only assists while you're pedaling, but it can help you reach speeds of up to 28 miles per hour. Because they're faster, Class 3 e-bikes often have additional rules depending on where you ride."

Discuss the three classes of e-bikes and explain that e-bikes travel faster than traditional bicycles, which means riders need more time and distance to stop safely.

Discuss the importance of:

- Wearing a properly fitted helmet
- Slowing down around pedestrians
- Following traffic laws
- Staying off phones while riding
- Charging batteries safely according to manufacturer instructions

Ask students:

"What could happen if someone rides an e-bike too fast around pedestrians?"

Allow students to share their responses and discuss why responsible riding helps keep everyone safe.

Lesson 1 Conclusion (10 Minutes)

Teacher Script

"Today we learned several important skills that every rider should know before getting on a bicycle or e-bike."

Ask students:

- "What's one thing you'll always check before riding?"
- "What's one difference between a bicycle and an e-bike?"

Have students complete an exit ticket before leaving class.

Lesson 2: Bike Skills Course (Outdoor)

Time: 60 Minutes

Introduction

Teacher Script

"Welcome back! During our last lesson, we learned about bicycle, pedestrian, and e-bike safety in the classroom. Today we'll practice those same skills by participating in a Bike Skills Course."

"Our goal today isn't to ride as fast as possible. Our goal is to ride safely, demonstrate good decision-making, and practice the skills we've learned."

Review safety expectations before students begin.

- Wear a properly fitted helmet at all times.
- Wait for the teacher's signal before starting.
- Ride at a safe, controlled speed—no racing.
- Keep a safe distance from other riders.
- Follow the course direction and obey all signs and cones.
- Use hand signals when instructed.
- Stay alert and watch where you're riding.
- If you need help or have a problem, stop safely and notify the instructor.

Station 1: Helmet Fit Check

Teacher Script.

"Before anyone begins riding, check your partner's helmet to make sure it fits correctly. Remember that the helmet should sit level on the head, the straps should form a 'V' under each ear, and the chin strap should be snug."

Allow students to complete helmet checks before moving to the next station.

Station 2: Hand Signal Practice

Teacher Script

"Before riding on the course, show me your left-turn signal, right-turn signal, and stop signal."

Provide feedback and have students repeat the signals until they demonstrate them correctly.

Station 3: Obstacle Course

Teacher Script

"Ride through the cone course while practicing smooth starts, controlled turns, safe stopping, and maintaining a safe speed."

Observe students and provide feedback on their riding skills throughout the activity.

Station 4: Traffic Sign Challenge

Teacher Script

"As you approach each traffic sign, stop and tell me what the sign means before continuing through the course."

Ask follow-up questions if needed to reinforce understanding.

Station 5: Safe Pedestrian Crossing

Teacher Script

"Imagine you've parked your bike and now need to cross the street. Show me how you would safely cross."

Students should stop, look left-right-left, make eye contact with drivers when appropriate, and cross safely.

Station 6: E-Bike Decision Challenge

Teacher Script

"I'm going to read different scenarios. After each one, decide whether the situation is safe or unsafe and explain why."

Example scenarios include:

- Riding an e-bike too fast on a crowded sidewalk (unsafe)
- Wearing a helmet while riding (safe)
- Using a phone while riding (unsafe)
- Carrying another passenger on an e-bike (unsafe)
- Slowing down near pedestrians (safe)
- Obeying traffic signs and signals (safe)

Encourage students to explain their reasoning and discuss why safe decision-making is important.

Lesson 2 Conclusion (10 Minutes)

Teacher Script

"Excellent job today! You practiced many of the same skills that experienced cyclists use every time they ride."

Ask students to share one new safety habit they plan to use the next time they walk, ride a bicycle, or ride an e-bike.

Conclude by saying:

"Remember, riding safely isn't just about protecting yourself—it also helps protect everyone around you. Whether you're walking, biking, or riding an e-bike, making safe choices helps create safer communities for everyone."